Capstone Expo

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Assess the explanation, trace and limits of the actual project.

Model supported examples, withhold the independent task, preserve support records and plan a delayed return.

Keep one habit: Require enough evidence to reproduce the choice; keep software correctness separate from unmeasured learning benefit.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Defend an application using reproducible **evidence**

Teach Week 36 as a two-session investigation using fictional data. Finish the fictional activity chooser with a short **evidence**-based presentation and a reproducible **handover**.

Show the purpose, contract, pipeline, one ordinary trace, one boundary or failure case, improvements and remaining limits. A working **demonstration** supports a software claim, not a claim of proven learning benefit.

Assess the three stated outcomes through the independent task, not by copying the worked result.

By the end, I can defend a complete recommendation with validation, eligibility and reproducible ranking.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Python 3 editor with a fresh local practice file
- Paper trace table and the student workbook
- Fictional **input** records from this lesson; no real student identifiers

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Improve and Challenge. Give an example and a mistake that the idea helps you avoid.

Look for: Repair a specific observed failure

Reference example, not the only acceptable wording: Strict less-than excludes equality; at most requires <=.

Without opening the old lesson, explain one key idea from Build the Recommender. Give an example and a mistake that the idea helps you avoid.

Look for: Implement a deterministic recommender

Reference example, not the only acceptable wording: Both score 8 and qualify; A is alphabetically first under the tie rule.

Without opening the old lesson, explain one key idea from Agents and Tools. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of state machines for bounded agents

Reference example, not the only acceptable wording: Teacher explanation: A bounded agent uses explicit states and finite action limits. A state machine makes invalid transitions visible.

States are ready, looking_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised **evidence**. done is terminal. Maximum two lookup attempts.

Identified **input**s, code/configuration, rules and expected **output**s that another person can rerun.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Improve and Challenge

Build the Recommender

Agents and Tools

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Present a recommender that gets 8/10 final requests right. One failure invents a title; another violates a hard page limit. Is '**80%**, ready for everyone' justified? Propose release criteria.

Concept to explain

Ownership is demonstrated by a fresh recommendation and **evidence**-based **defense**.

Term	Meaning
capstone	A final project combining skills and evidence developed across the course.
defense	An explanation of a project decision supported by evidence and an account of its limitations.
transfer	Apply an idea or skill to a new case rather than copy the worked example.
limitation	A condition or missing capability that restricts what a result can establish or a system can safely do.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked **demonstration** 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Model a reproducible defence rather than a winner-only demo

> Keep illustrative activity report and supported book catalogue separate.

State the handover purpose

Finish the fictional activity chooser with a short **evidence**-based presentation and a reproducible **handover**.

Show the purpose, contract, pipeline, one ordinary trace, one boundary or failure case, improvements and remaining limits. A working **demonstration** supports a software claim, not a claim of proven learning benefit.

Run the small count-consistency check

```
report = {"supplied": 4, "accepted": 3, "rejected": 1,
          "eligible": 2, "selected": "A", "score": 8}
assert report["accepted"] + report["rejected"] == report["supplied"]
assert report["eligible"] <= report["accepted"]
print(report["selected"], report["score"])
```

The example report prints A 8 and its counts are internally consistent. These checks do not prove A is the correct winner; the **input** records and scoring trace are also needed.

Keep raw fictional data, source code, settings, test results and change log together.

> **A demonstration shows only the winning name. Name two pieces of evidence needed to explain that choice.**

Read the pictured practice catalogue

A fictional team presents its transparent book recommender on a supplied practice catalogue.

Request maximum 120 pages; score=2×interest+illustrated, where illustrated is 1 for True and 0 for False; tie by ID.

A pages120 interest3 True; B pages100 interest4 False; C pages121 interest5 True; D pages90 interest"4" True. Numeric fields require exact integers.

Trace validation, filtering and scores



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Validate	D has text interest and is rejected; A,B,C have valid fields.
2. Filter	A and B meet the page limit; C exceeds it.
3. Rank	A scores 7; B scores 8, so B wins.
4. Defend evidence	Explain two eligible items, one page-limit exclusion and one type rejection.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A presentation should let someone rebuild the result from the stated contract. A's equality is accepted; D is not silently converted; C's high preference score cannot override the page requirement. Reversing rows must preserve B because it has the unique highest eligible score. This pictured case is supported practice; keep the separately named independent activity task for a first attempt before feedback.

Predict the authorised input correction

Change: An authorised repair converts D's interest to integer 4.

Prediction: D becomes eligible and scores 9, so it wins.

Reason: The **evidence**-backed **input** repair changes validation and then ranking; record this as a new case/version.

Run the complete pictured book program

```
from collections import Counter

def book_report(raw, limit):
    if type(limit) is not int or limit < 0:
        return {'error':'invalid page limit'}
    names = Counter(r['id'] for r in raw if type(r) is dict
                    and type(r.get('id')) is str and r['id'])
    accepted, rejected = [], []
    for index, row in enumerate(raw,1):
        reason = None
        if type(row) is not dict:
            reason = 'not a record'
        elif type(row.get('id')) is not str or not row['id']:
            reason = 'invalid ID'
        elif names[row['id']] > 1:
            reason = 'duplicate ID'
        elif type(row.get('pages')) is not int or row['pages'] < 0:
            reason = 'invalid pages'
        elif type(row.get('interest')) is not int or not 0 <= row['interest'] <= 5:
            reason = 'invalid interest'
        elif type(row.get('illustrated')) is not bool:
            reason = 'invalid illustrated flag'
        if reason:
            rejected.append((index,reason))
        else:
            accepted.append(dict(row))
    eligible = [r for r in accepted if r['pages']<=limit]
    ranked = sorted(eligible, key=lambda r: (-
(2*r['interest']+int(r['illustrated'])),r['id']))
    scores = [(r['id'],2*r['interest']+int(r['illustrated'])) for r in ranked]
    return dict(supplied=len(raw),accepted=len(accepted),rejected=rejected,
                eligible=len(eligible),ranked=scores,
                selected=scores[0][0] if scores else None,
                over_limit=[r['id'] for r in accepted if r['pages']>limit])

books = [dict(id='A',pages=120,interest=3,illustrated=True),
         dict(id='B',pages=100,interest=4,illustrated=False),
         dict(id='C',pages=121,interest=5,illustrated=True),
         dict(id='D',pages=90,interest='4',illustrated=True)]
original = book_report(books,120)
print(original)
# Explicitly supplied authorised toy-data correction; keep the original unchanged.
changed = [dict(r,interest=4) if r['id']=='D' else dict(r) for r in books]
```

```
print(book_report(changed,120))
assert book_report(list(reversed(books)),120)['ranked']==original['ranked']
```

State evidence limits and preserve independent assessment

A defensible algorithmic trace does not prove B will be liked or that classroom learning improved.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Demo one known reader profile end to end.

2. Recalculate one book score in front of the audience.

3. Accept an unseen profile and predict before running.

4. State one **limitation**, one fairness check, and one responsible next improvement.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

An expo claim says the recommender knows everybody's favourite book after twelve tests.

Rewrite the defence.

Exact explanation to model

Explain trait-based ranking, report the twelve outcomes and limits, and demonstrate no-match handling. It cannot know unstated preferences.

Explain the added worked example

Key terms

Term	Meaning
Defense	An explanation supported by evidence and questions
Evidence bundle	The records needed to check a project claim
Demonstration	Showing the application on specified inputs
Handover	Giving another person enough information to use or check the work

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

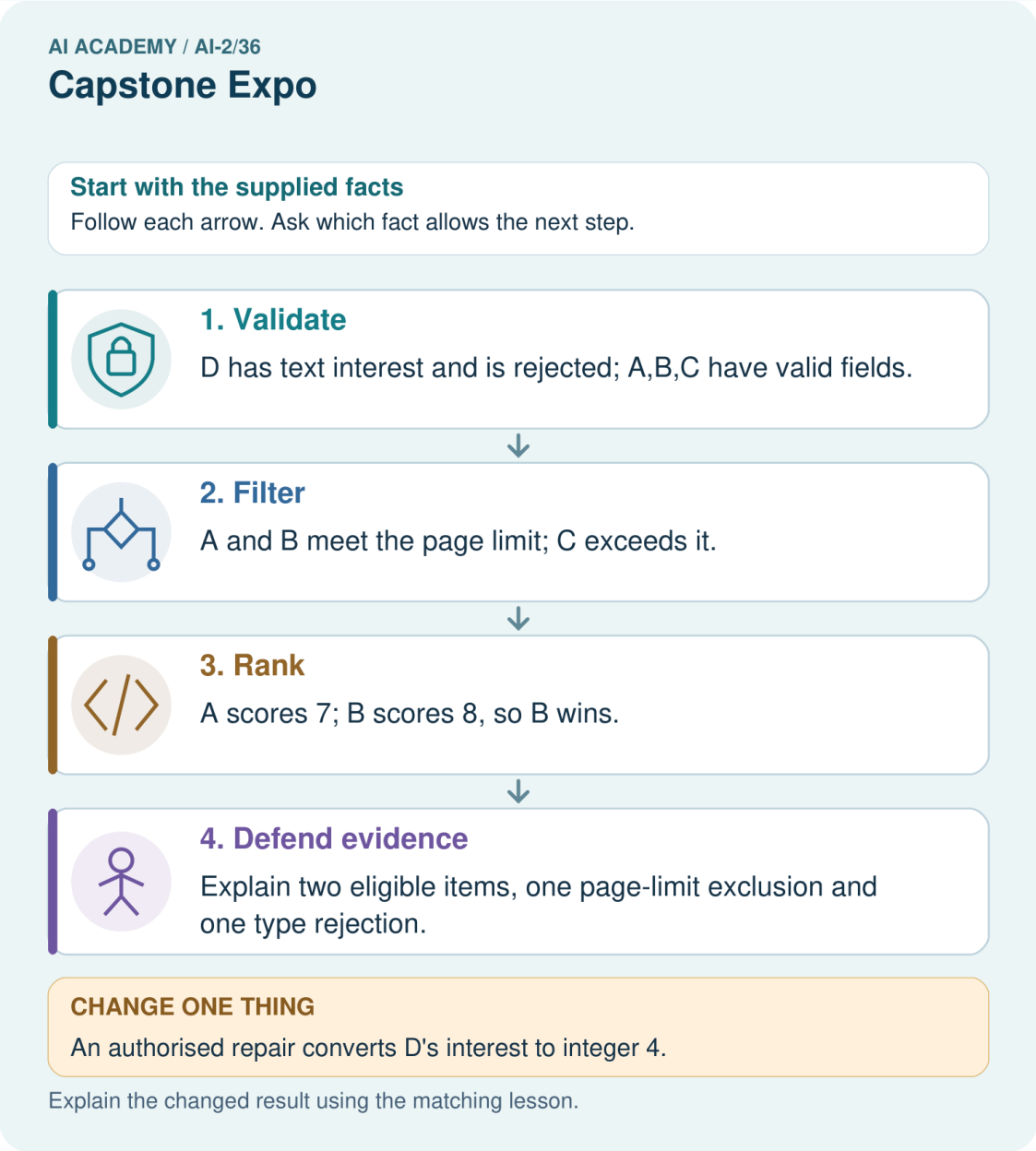
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Capstone Expo

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: A report says supplied 5, accepted 4 and rejected 2. Identify the inconsistency.

Expected reasoning and fading help

Teacher check: Accepted plus rejected is 6, not 5. Investigate double counting or reporting before presenting it.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for Defending the data application. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The final independent task must be attempted before hints. Record support honestly and use a new case after support.

Keep teacher answers separate from student materials and do not publish private credentials or real student records in the **handover**.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictorial check

Give the original winning ID, its score, and counts of eligible, excluded and invalid rows.

B wins with 8. Two rows eligible, one valid-but-too-long row excluded, one type-invalid row rejected.

Changed-case reasoning: D becomes eligible and scores 9, so it wins. The **evidence**-backed **input** repair changes validation and then ranking; record this as a new case/version.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ **Earlier wording, alternatives and picture transcripts**

Demonstrate and guide practice

Board sequence

Write the given facts: An expo claim says the recommender knows everybody's favourite book after twelve tests.

Write the question to explain: Rewrite the defence.

Reveal after **prediction**s: Explain trait-based ranking, report the twelve outcomes and limits, and demonstrate no-match handling. It cannot know unstated preferences.

Use the earlier changed question as optional later practice. Reserve Defend a new activity case independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

The **input** records and eligibility/scoring/tie trace; also show the policy/version and rejection counts. Exit: All possible **input** correctness and educational benefit remain unproven.

Add varied independent cases or propose a teacher-approved usability pilot with a defined question; do not claim outcomes before collecting **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.

Stage	Teacher action
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

Pictured book report: 4 supplied, 3 accepted, D rejected for text interest; A and B eligible, C over the limit. A scores 7 and B 8, so B wins. The explicit corrected-D version accepts 4 rows and has 3 eligible; D scores 9 and wins. Reversal preserves ranking; limit 0 has no eligible item. The small A/8 report demonstrates count consistency only.

Independent activity case: A and B score 8 and tie by name to A; C is valid but too long; D has invalid fun. Counts are 4 supplied, 3 accepted, 1 rejected, 2 eligible. Do not use the supported picture as fresh assessment.

Run and record

1. Predict → Test → Record → Explain

2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.

3. Rehearse with the complete pictured book program and fictional rows, preserving expected/actual outputs and the handover bundle. Keep the independent activity task withheld for a first attempt. A familiar rehearsal case cannot establish fresh transfer.

4. What did the hardest test teach you that the normal test did not?

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

A successful run proves the conclusion: The final independent task must be attempted before hints. Record support honestly and use a new case after support.

Keep teacher answers separate from student materials and do not publish private credentials or real student records in the **handover**.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry after support: limit 12; M (minutes 12, fun 3, learning 3), N (11,4,1), O (13,5,5), P (10,-1,2).

Counts 4 supplied,3 accepted,1 rejected,2 eligible. M/N tie at 9; M wins in either order.

A convenient shortcut preserves meaning: A polished presentation can conceal missing **evidence**. Replace claims such as works for everyone with the exact requirements met, cases executed and known **limitation**s.

Show one prior failure and its repair rather than deleting it from the story.

Choose support from the response

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Defend a new activity case independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: Limit is 10. A has minutes 10, fun 2, learning 4; B has 9,3,2; C has 11,5,5; D has 8,"4",2. Validate, rank, give counts, explain the winner, then reverse the **input** order. Present without hints.

4 supplied, 3 accepted, 1 rejected D for rating type, 2 eligible A/B. Both score 8; A wins alphabetically. C is valid but over-limit. Reversal preserves counts and A. Explain the policy and its limits.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Assemble evidence

List the files or records a peer needs to reproduce the chooser result.

Fictional raw dataset, schema, code version, settings, run instructions, tests with expected/actual results, rejection reasons and change log.

W2 Check report consistency

A report says supplied 5, accepted 4 and rejected 2. Identify the inconsistency.

Accepted plus rejected is 6, not 5. Investigate double counting or reporting before presenting it.

W3 Repair a claim

A presentation says the project proves students learn better. What claim does a successful rule-based demo support instead?

It supports that the tested version follows specified selection rules on the tested cases. Learning benefit needs separate classroom **evidence**.

W5 State the next evidence need

After passing the new case, what remains unproven and what is a reasonable next step?

All possible **input** correctness and educational benefit remain unproven. Add varied independent cases or propose a teacher-approved usability pilot with a defined question; do not claim outcomes before collecting **evidence**.

▼ **Earlier wording, alternatives and picture transcripts**

Assess and explain the answers

Worked case answer

Explain trait-based ranking, report the twelve outcomes and limits, and demonstrate no-match handling. It cannot know unstated preferences.

Additional transfer practice question

Your demo recommends the right book once. An audience member asks why they should trust it. Name **evidence** and one **limitation** you should present.

Reasoned practice answer

Show the scoring rule, test preferences, expected and observed **output**s, failures and revisions. Explain the limited test coverage and when no match or human review is needed.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A successful live example replaces an evaluation record.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

No. Report 8/10 on the specified set, but invented titles and constraint violations fail essential criteria. Repair and retest those behaviours, preserve a no-match path and gather broader **evidence**.

The system ranks declared traits; it does not know unstated preferences.

Explanations for the additional workbook examples

W1 Assemble evidence

W2 Check report consistency

W3 Repair a claim

W4 Defend a new case independently

Limit is 10. A has minutes 10, fun 2, learning 4; B has 9,3,2; C has 11,5,5; D has 8,"4",2. Validate, rank, give counts, explain the winner, then reverse the **input** order. Present without hints.

4 supplied, 3 accepted, 1 rejected D for rating type, 2 eligible A/B. Both score 8; A wins alphabetically. C is valid but over-limit. Reversal preserves counts and A. Explain the policy and its limits.

W5 State the next evidence need

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Final **evidence bundle**: brief, data, scoring logic, tests, improvement record, and product card.

Keep the decision, **input**, result, explanation and remaining **limitation** in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Present the Transparent Book Recommender's code, baseline, full-case traces, reserved results and **limitation**s, without claiming fictional results as pupil work.

Support while preserving the reasoning goal

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Offer a presentation outline only after the initial independent attempt and record support. Extension: let a peer supply a new valid case and require a **prediction** before running.

Assess explanation, not decorative slide polish.

Independent W4 **evidence**: 4 supplied, 3 accepted, 1 rejected D for rating type, 2 eligible A/B. Both score 8; A wins alphabetically. C is valid but over-limit. Reversal preserves counts and A. Explain the policy and its limits.

Record whether the learner needed no help, a hint or explanation. Fresh retry after support: limit 12; M (minutes 12, fun 3, learning 3), N (11,4,1), O (13,5,5), P (10,-1,2). Counts 4 supplied,3 accepted,1 rejected,2 eligible.

M/N tie at 9; M wins in either order.

Plan the delayed check

Suggested return points in this level: plan a review after the level ends. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous **capstone** before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	

Record	Teacher evidence
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Review the full project **evidence** and choose concepts needing another independent attempt before the next level. One expo is not proof of mastery of all AI.

After Level2, continue to Level3: Generative AI Systems. Revisit the project's explicit contracts, validation and reproducibility before starting Predictive and Generative AI in the next level; use recorded readiness and support **evidence** rather than expo polish as the progression decision.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Transparent Book Recommender

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Defend an application using reproducible **evidence**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: A report says supplied 5, accepted 4 and rejected 2. Identify the inconsistency.

Reasoned practice answer: Accepted plus rejected is 6, not 5. Investigate double counting or reporting before presenting it.

Check every learner before discussion: Use the worked case for Defending the data application. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The final independent task must be attempted before hints. Record support honestly and use a new case after support.

Keep teacher answers separate from student materials and do not publish private credentials or real student records in the **handover**.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: An authorised repair converts D’s interest to integer 4. Require a **prediction** and a reason before comparing. Expected result: D becomes eligible and scores 9, so it wins.

Reason: The **evidence**-backed **input** repair changes validation and then ranking; record this as a new case/version.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and **limitation**s. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

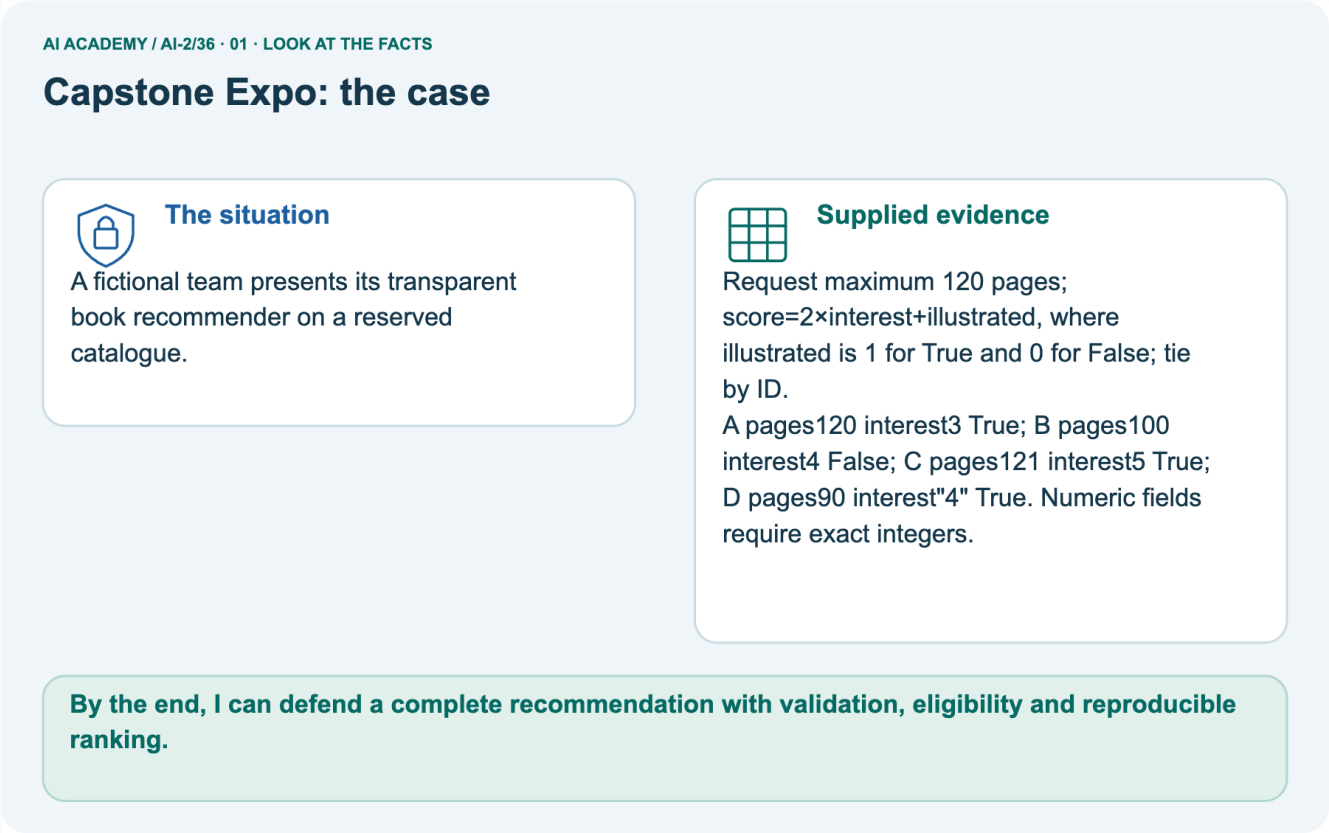
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Capstone Expo: the case



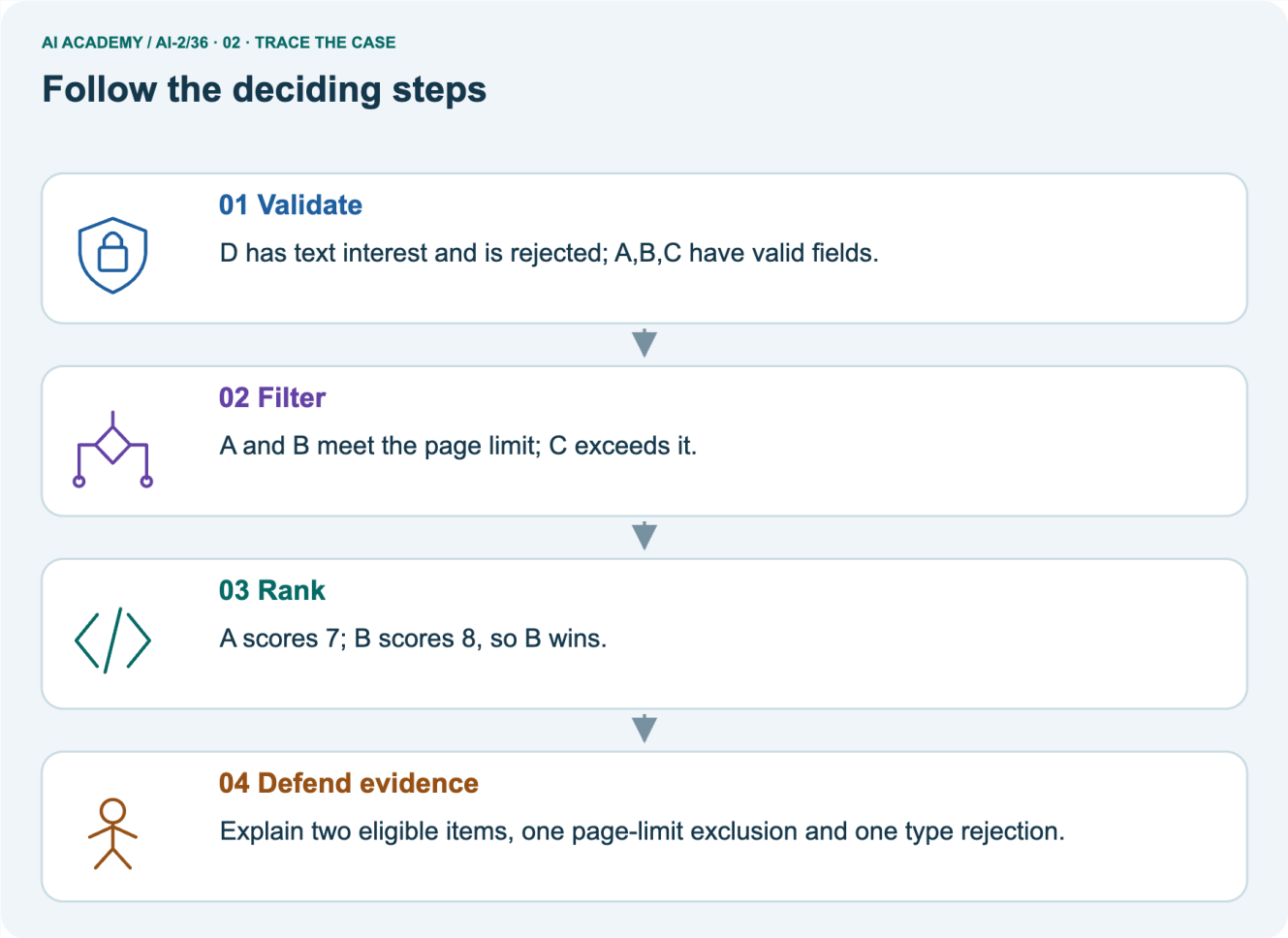
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional team presents its transparent book recommender on a supplied practice catalogue. Supplied facts: Request maximum 120 pages; score=2×interest+illustrated, where illustrated is 1 for True and 0 for False; tie by ID. A pages120 interest3 True; B pages100 interest4 False; C pages121 interest5 True; D pages90 interest"4" True. Numeric fields require exact integers.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



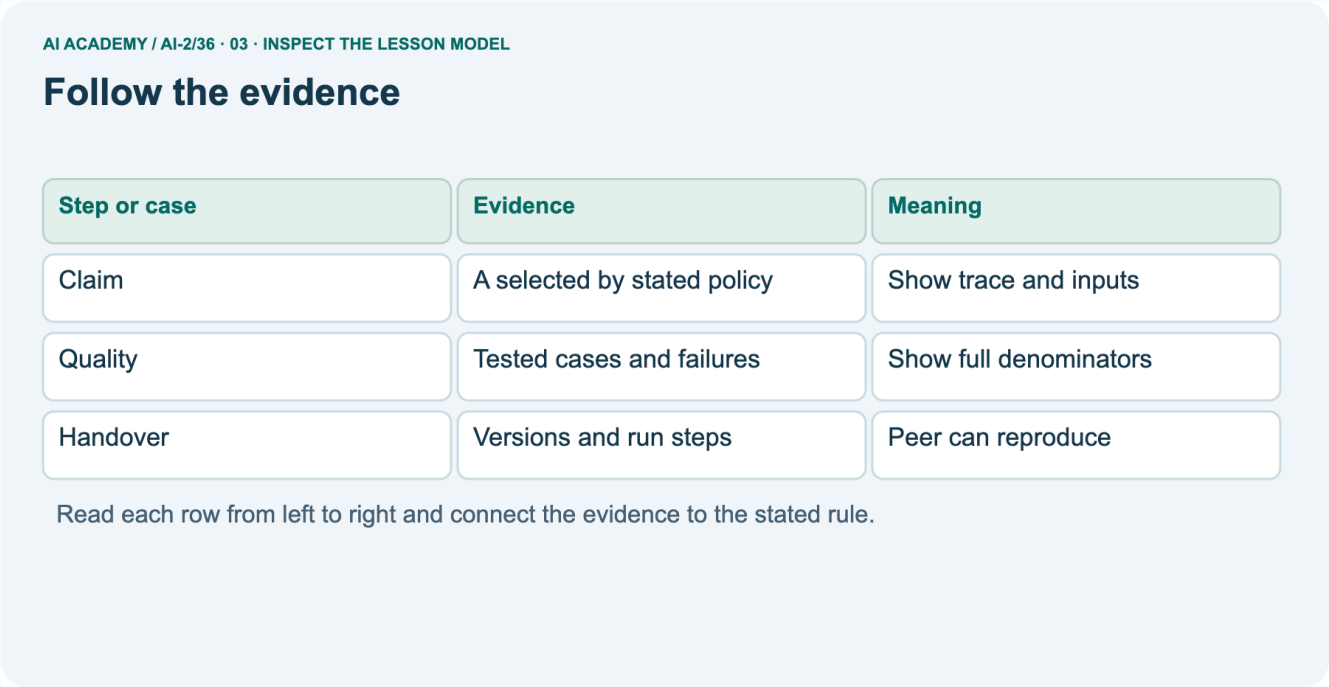
Follow the labelled picture with the matching explanation. Open the vector diagram

Validate: D has text interest and is rejected; A,B,C have valid fields. → Filter: A and B meet the page limit; C exceeds it. → Rank: A scores 7; B scores 8, so B wins. → Defend **evidence**: Explain two eligible items, one page-limit exclusion and one type rejection.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Follow the evidence



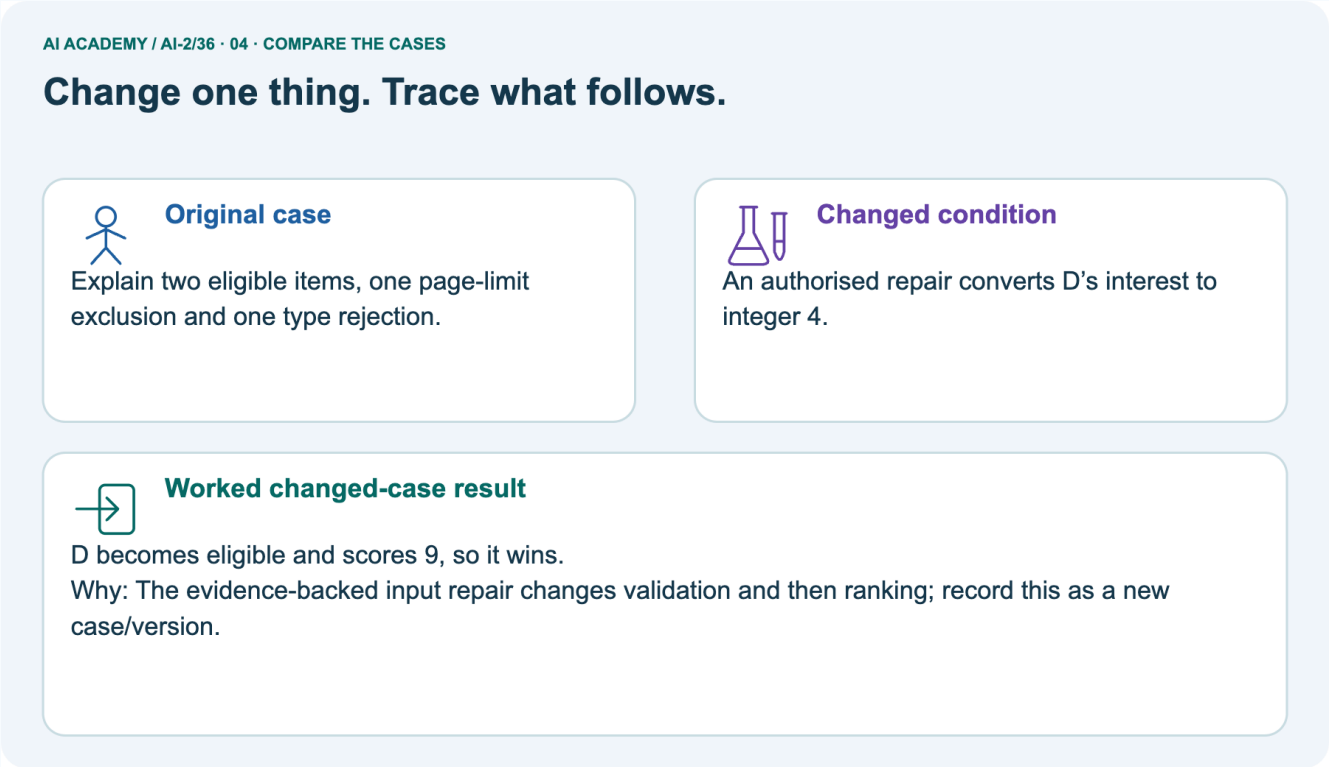
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Claim | A selected by stated policy | Show trace and **input**s. Quality | Tested cases and failures | Show full denominators. **Handover** | Versions and run steps | Peer can reproduce

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Explain two eligible items, one page-limit exclusion and one type rejection. Changed condition: An authorised repair converts D's interest to integer 4.

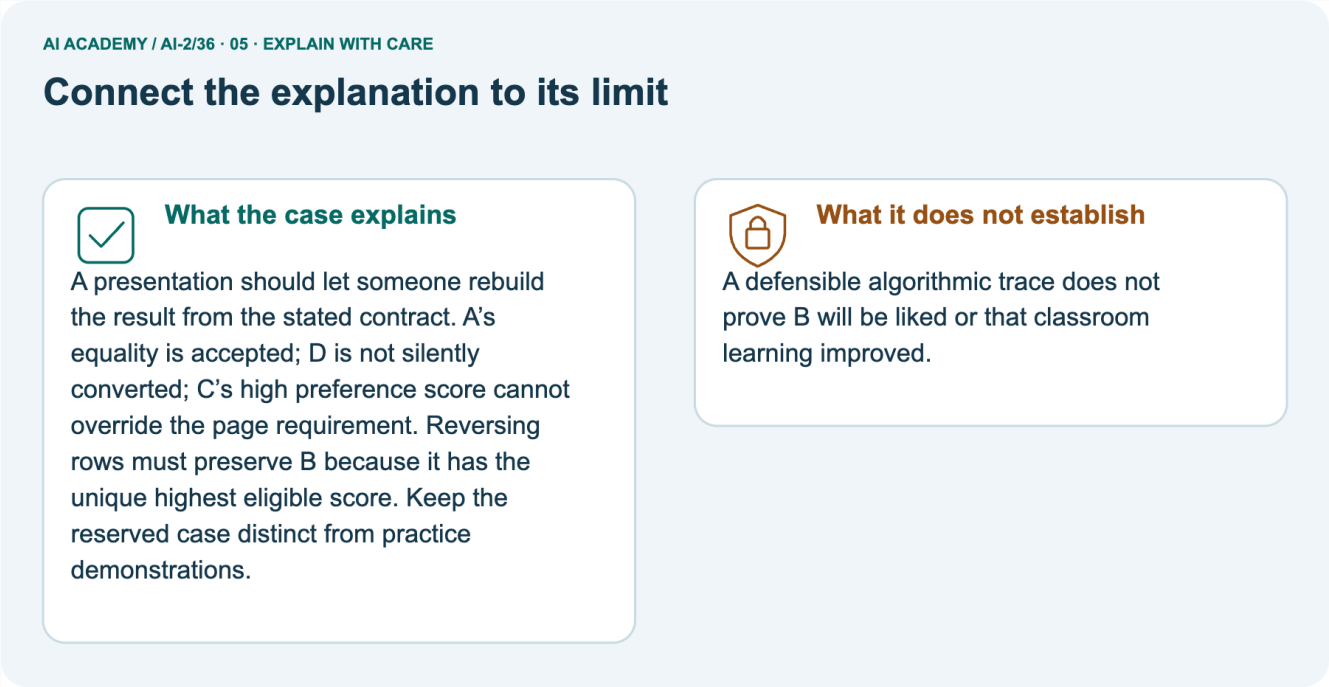
Expected result: D becomes eligible and scores 9, so it wins. Reason: The **evidence**-backed **input** repair changes validation and then ranking; record this as a new case/version.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The **evidence**-backed **input** repair changes validation and then ranking; record this as a new case/version.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

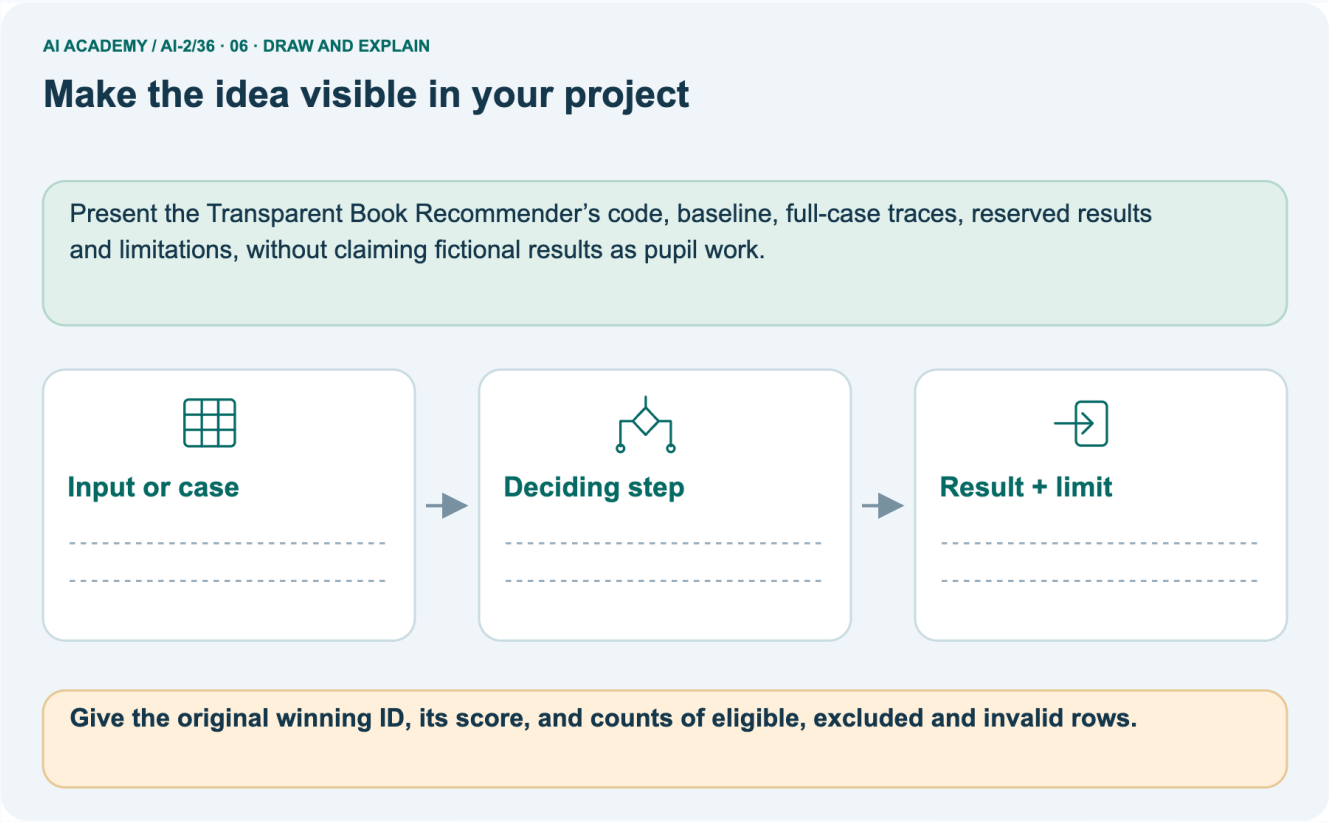
A presentation should let someone rebuild the result from the stated contract. A's equality is accepted; D is not silently converted; C's high preference score cannot override the page requirement. Reversing rows must preserve B because it has the unique highest eligible score. This pictured case is supported practice; keep the separately named independent activity task for a first attempt before feedback. Limit: A defensible algorithmic trace does not prove B will be liked or that classroom learning improved.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A defensible algorithmic trace does not prove B will be liked or that classroom learning improved.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Present the Transparent Book Recommender's code, baseline, full-case traces, reserved results and **limitation**s, without claiming fictional results as pupil work. Draw your **input**, deciding step, result and limit.

Supported question: Give the original winning ID, its score, and counts of eligible, excluded and invalid rows.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: B wins with 8. Two rows eligible, one valid-but-too-long row excluded, one type-invalid row rejected.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Capstone Expo: the case

02. Follow the deciding steps

03. Follow the evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.